

Optimization of plate-fin heat sink configurations for enhanced thermal performance and manufacturability

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ABSTRACT

Enhancing heat sink efficiency presents a significant challenge, requiring the optimization of heat transfer performance while minimizing pressure drop across the inlet and outlet. Although previous designs have improved heat sink performance, their complex geometries have resulted in high manufacturing costs. This study introduces four novel plate-fin heat sink configurations—fillet, chamfer, step, and concave fillet—designed for enhanced manufacturability. A conjugate heat transfer model was employed to analyze forced convection heat transfer over a Reynolds number (Re) range of 500–5000, with laminar and turbulence models validated against experimental data to ensure accuracy near the interface surface. The results indicate that the $k-\omega$ turbulence model achieved excellent predictive accuracy, with an average experimental error of less than 5.07 %. Moreover, the fillet, chamfer, step, and concave fillet plate-fin heat sinks exhibited thermal enhancement efficiencies exceeding those of conventional designs at the $Re = 5000$ by 17.3 %, 15.9 %, 0.8 and 4.6 %, respectively. However, the step plate-fin heat sink did not yield thermal performance improvements despite a lower friction factor than the conventional design. To support future heat sink development, the optimized t/R and t/C ratios were determined to be 2.0 and 1.2 for the fillet and chamfer plate-fin heat sinks, facilitating maximum enhancement of both design and manufacturing processes.

1. Introduction

A heat sink is a critical thermal management device designed to dissipate heat from electronic components, preventing overheating and ensuring operational stability. Forced convection serves as the primary heat transfer mechanism in heat sinks, facilitating efficient thermal dissipation [1,2]. The effectiveness of a heat sink is typically assessed through key parameters such as the heat transfer

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coefficient, Nusselt number, and thermal enhancement efficiency, which collectively determine its cooling performance [3–8]. Furthermore, heat sinks can be fabricated using various manufacturing techniques, including machining, forging, stamping, extrusion, casting, folding, skiving, and 3D printing, with the choice of method significantly influencing both the geometric design and production feasibility [9–11]. Given the direct impact of manufacturing costs on the overall cost-effectiveness of heat sink production, optimizing both thermal performance and economic feasibility remains a fundamental challenge in heat sink design and development.

The plate-fin heat sink is a widely studied type of heat sink due to its versatility in manufacturing cost reduction. Consequently, it has been extensively designed and developed in previous research. Lan et al. [12] utilized computational fluid dynamics (CFD) to examine the thermal performance (Nusselt number) and hydraulic characteristics (friction factor) of a plate-fin heat sink. Their simulation results were validated against experimental data. Additionally, they analyzed the effects of varying Reynolds numbers, fin widths, and fin counts to identify the optimal plate-fin heat sink configuration. A genetic algorithm-backpropagation (GA-BP) approach was employed for optimization. Boujelbene et al. [13] investigated the influence of inclination angles on the performance of heat sinks, focusing on tree-shaped and plate-fin configurations. Among the designs evaluated, the plate-fin and tree-shaped heat sinks with metal volume fractions of 20 % and 30 %, respectively, exhibited the highest and lowest temperature non-uniformity within the metal. Nilpueng et al. [14] incorporated copper foam and twisted tapes to improve the heat transfer performance of plate-fin heat sinks. Lan et al. [15] varied bionic-fin lengths, widths, and heights for analysis of the thermal properties with CFD, as well as plate-fin heat sinks positioned at different locations. The results showed that at a Reynolds number of 6000, the thermal enhancement efficiency of the bionic rib decreased by up to 15 % as rib height increased. In contrast, increasing rib length improved efficiency by up to 10.38 %, while increasing rib width led to a reduction of up to 2.07 %. Regarding placement, the bionic rib was most effective in enhancing thermal performance when positioned near the channel outlet. Wang et al. [16] developed a double L-shaped heat sink with a micro heat pipe array and copper foam wick for high-power electronics cooling. Experiments and simulations assessed heat transfer factor under varying copper foam lengths, inclination angles, and liquid filling rates. Results showed that the composite wick structure reduced thermal resistance by 23 % to 0.258 K/W. Fin structure optimization stabilized a 214 W chip at 70.1 °C, lowering its temperature by 8.7 % from the pre-optimization state. Mehjabeen et al. [17] employed CFD simulations to compare three novel heat sink designs—arrowhead, concave plate, and hexagonal plate-fin—with a conventional rectangular plate-fin heat sink. The results indicated that the hexagonal plate-fin heat sink with elliptical perforations achieved the highest hydrothermal performance factor. Selvan et al. [18] optimized a tapered plate-fin heat sink for a bus bar conductor. The optimal design featured six fins with a pitch of 4 mm, a height of 6.5 mm, and a thickness of 1 mm. This configuration resulted in an average surface temperature of 72.05 °C and a corresponding Nusselt number of 21.59. Chowdhury et al. [19] modified the pin design of plate pin fin heat sinks (PPFHSs) using CFD simulations (ANSYS 2021 R2) to enhance performance while reducing power consumption. The PPFHS-Rook design achieved the highest thermal performance factor (TPF), improving by 3.2 % over the original PPFHS. It also exhibited the lowest pressure drop, reduced by 12.849 %. Arshad [20] used CFD simulations (ANSYS-FLUENT 23 R1) to analyze the heat transfer performance of a novel

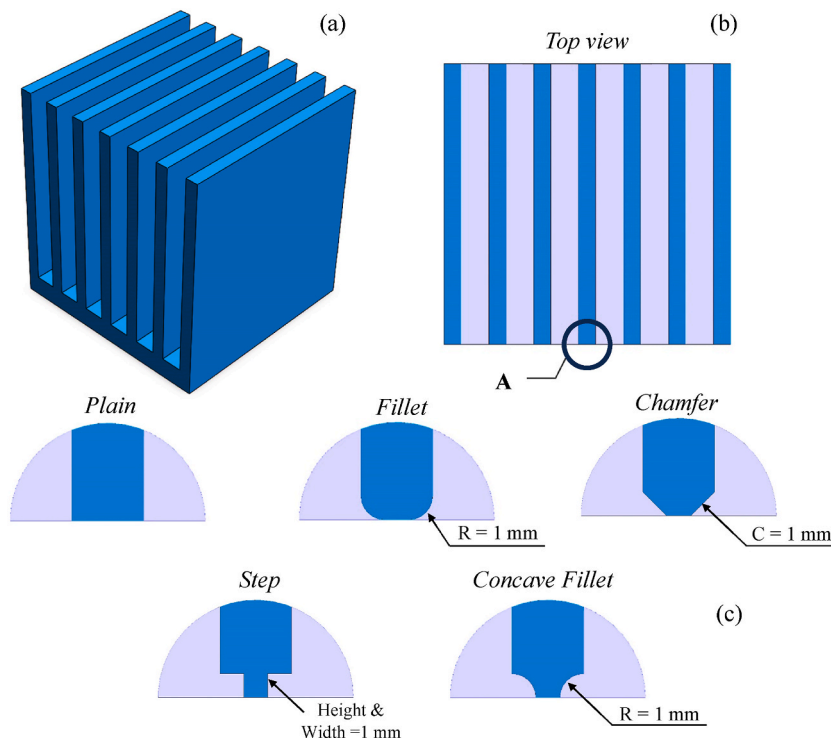


Fig. 1. (a) A conventional plate-fin heat sink and (b) the proposed plate-fin configurations, highlighting different inlet modifications.

tree-shaped fin heat sink integrated with phase change material. Ji et al. [21] examined how the twist angle and variations in NACA airfoil cross-sections influence the flow and heat transfer characteristics of blade-like fin heat sinks. Using numerical simulations across a Reynolds number range of 8,548 to 34,194, their study revealed that the innovative twisted blade-like fins achieved significantly better overall thermal performance compared to conventional cylindrical fins. Noui et al. [22] utilized a 3D conjugate heat transfer model to investigate how geometric configurations and hydrodynamic parameters influence thermal resistance and pumping power in mini-channel heat sinks. The converging-diverging mini-channel configuration, using metal Galinstan, demonstrated strong performance in exergy evaluations and achieved a more uniform temperature distribution, with a maximum temperature of approximately 328 K. Marinaro et al. [23] employed topology optimization to design forced air-cooled heat sinks for an inverter used in aeronautical propulsion systems. In aircraft applications, heat sink design must balance multiple factors, including minimizing thermal resistance and fan power consumption while also reducing weight and volume.

Achieving a high heat transfer coefficient while minimizing the pressure drop between the inlet and outlet is essential for optimizing the thermal performance factor of plate-fin heat sinks. Previous studies have explored performance enhancements through modifications such as altering rectangular plate fins and incorporating pin fins or porous materials. While effective in improving thermal performance, these methods often increase material consumption and introduce manufacturing complexities. To address these challenges, this study proposes an innovative optimization approach that enhances plate-fin heat sink performance without requiring additional material or intricate fabrication techniques. CFD simulations were employed to develop and validate the proposed method, with numerical results benchmarked against experimental data from [24]. A suitable turbulence model was implemented to accurately capture conjugate heat transfer effects, ensuring reliable thermal and flow predictions. The proposed design was systematically compared with conventional and geometrically complex configurations to assess its effectiveness, followed by an optimization process to maximize performance. The findings demonstrate that this approach offers a practical and efficient solution for advancing plate-fin heat sink design, with broad applicability in thermal management systems.

2. Numerical approaches

2.1. Plate-fin designs

A conventional heat sink, as shown in Fig. 1(a), is utilized as the benchmark model. It has been observed that minor modifications to this conventional configuration can enhance the heat transfer coefficient while minimizing the different pressure values between the

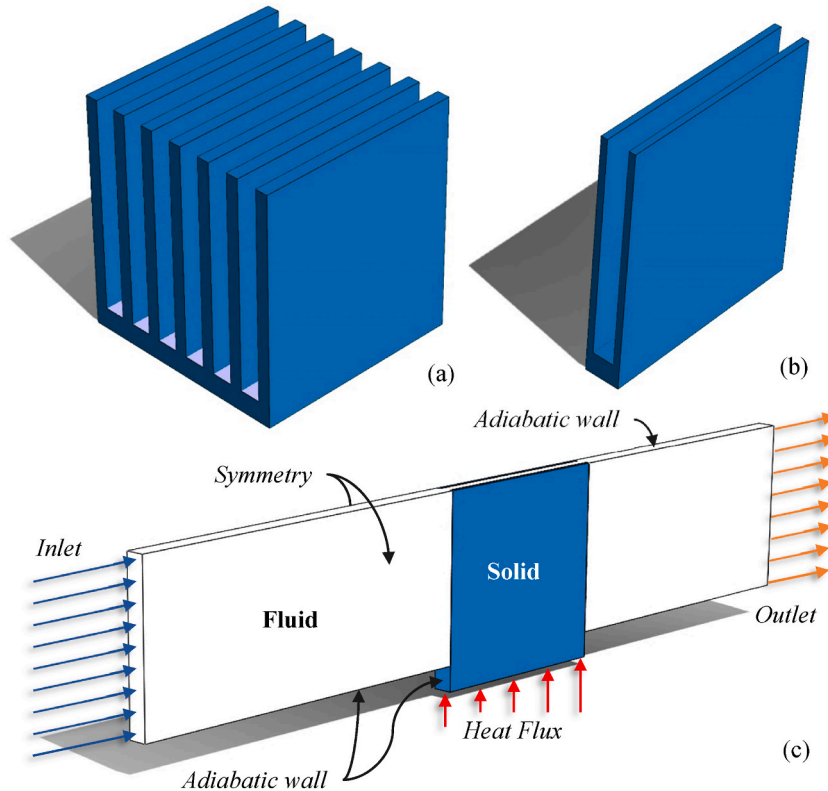


Fig. 2. Computational models illustrating (a) the conventional heat sink, (b) a single heat sink channel, and (c) the computational domain for conjugate forced convective heat transfer around the channel heat sink.

inlet and outlet. Specifically, reducing the sharp corners of the heat sink fins at the inlet has been identified as an effective improvement. Consequently, four novel plate-fin heat sink designs—fillet, chamfer, step, and concave fillet—are proposed, as illustrated in Fig. 2(b), to investigate factors influencing heat transfer enhancement. These design modifications are not only simple and cost-effective but also easier to manufacture compared to previously developed complex heat sink configurations [22,25–28]. Although more intricate designs, such as bionic leaf-vein-like heat sinks [25] or porous-fin heat sinks [40,41], have demonstrated improved thermal performance, they typically involve higher manufacturing complexity and cost.

Plate-fin heat sinks can be produced through extrusion, a process where metal is shaped by being pushed through a die. This method is widely used due to its relatively low tooling and per-unit costs. Table 1 outlines the production costs and characteristics of various manufacturing techniques [29]. After the extrusion step, additional machining—such as the addition of fillets, chamfers, steps, and concave fillets at the inlet—is performed, which introduces a modest increase in cost. Even with these enhancements, the proposed plate-fin heat sink remains more cost-effective than designs involving more intricate fin structures, which require higher-cost processes like bonding, folding, forging, skiving, or full machining.

2.2. Numerical models

A conjugate heat transfer (CHT) approach was employed to simulate steady-state forced convection through a plate-fin heat sink, capturing the coupled thermal interactions between the solid and fluid domains. The simulations were conducted under turbulent flow conditions with fixed boundary conditions and executed as single-run computations. As a result, no statistical averaging was performed, and error bars are not presented, since the model does not incorporate variability or multiple trials. Within the fluid domain, the governing equations of continuity, momentum, and energy were solved in their conservative form. These equations are presented as follows [30–33]:

Fluid:

Continuity equation:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \bar{U}) = 0 \quad (1)$$

Momentum equation:

$$\frac{\partial}{\partial t} (\rho \bar{U}) + \nabla \cdot (\rho \bar{U} \bar{U}) = -\nabla \bar{p} + \nu \nabla \cdot (\nabla \bar{U}) - \rho \nabla \bar{U} \bar{U} + S_{\bar{U}} \quad (2)$$

Energy equation:

$$\frac{\partial \rho}{\partial t} (\rho \bar{T}) + \nabla \cdot (\rho \bar{U} \bar{T}) = \left(\frac{\nu}{Pr} \right) \nabla \cdot (\nabla \bar{T}) - \rho \nabla \bar{T} \bar{U} + S_{\bar{T}} \quad (3)$$

Solid:

$$\frac{\partial \rho}{\partial t} (\rho C_p \bar{T}) = \nabla \cdot (\lambda \cdot \nabla \bar{T}) + S_{\bar{T}} \quad (4)$$

In these equations, the following variables are defined: t represents time, ρ is the density of the fluid and solid, $\bar{U}$ denotes average velocity, $\bar{p}$ denotes the average pressure, ν is the kinematic viscosity, Pr is the Prandtl number, $\rho \bar{U} \bar{U}$ is the Reynolds stress term, $\rho \bar{T} \bar{U}$ denotes the turbulent heat flux, λ denotes thermal conductivity, C_p is the heat capacity at constant pressure, $S_{\bar{U}}$ denotes the momentum source term influenced by average velocity and $S_{\bar{T}}$ is the energy source term influenced by temperature.

In this study, the interface between the solid and fluid domains was assumed to be perfectly bonded, with no interfacial thermal resistance. This assumption is commonly employed in conjugate heat transfer simulations using ANSYS 2022 R2, wherein continuity of both temperature and heat flux is enforced at the interface. Although contact resistance can influence heat transfer in practical applications, it was excluded from the present model to focus on idealized thermal performance. This condition implies that the heat flux entering one region from the interface must be balanced by the heat flux leaving the other region on the opposite side, expressed as:

$$Q_{fluid} = -Q_{solid} \quad (5)$$

To solve the governing equations, the SIMPLE algorithm was employed to address the pressure-velocity coupling problem.

2.3. Heat sink model configuration

Conventional plate-fin heat sinks provide an effective thermal management solution by incorporating multiple parallel channels, as illustrated in Fig. 2(a). The heat sink design follows specific dimensions: a width and length of $0.05 \text{ m} \times 0.05 \text{ m}$, a channel height of 0.05 m , a channel width of 0.005 m , a wall thickness of 0.003 m , and a base thickness of 0.02 m . The plate fins are securely attached to the base, which connects the inlet and outlet sections.

To develop the computational model, this study evaluates the performance of a simplified, symmetry-based approach. This choice is justified by the availability of extensively validated full three-dimensional simulations based on the same benchmark dataset [2,24], which enables an assessment of whether the reduced model can accurately capture the essential flow dynamics while significantly

Table 1
Characteristics and cost comparison of heat sink manufacturing methods.

Method	Properties	Spacing ratio	Cost
Extrusion	Most common approach. Low overall cost. Limited to low aspect ratios.	Low	\$
Die-casting	Fast production. Requires high upfront investment. Lower thermal performance than extrusion.	Low	\$
Stamping	Suited for high-volume production. Typically used for low-performance applications.	N/A	\$
Bonding	Expensive. Thermal and mechanical reliability depend on the thermal paste used.	High	\$\$\$
Folding	Design flexibility. Lightweight. Some contact resistance due to soldering/brazing. Challenging to achieve tight fin spacing.	Medium/High	\$\$\$
Forging	Produces strong, rigid, and uniform fin structures.	High	\$
Skiving	Allows high fin density with minimal tooling. Limited height/pitch for copper materials.	Medium	\$\$
Machining	Prone to fin damage. Material waste is significant.	High	\$\$\$

reducing computational cost. As shown in Fig. 2(b), optimizing channel geometry, fluid flow characteristics, pressure drop, and thermal resistance is essential for enhancing heat dissipation efficiency, particularly in modern electronics. Each parallel channel acts as an independent pathway for fluid movement, ensuring efficient heat transfer [23]. Additionally, the symmetric and periodic structure of the heat sink reduces computational time in numerical simulations [3,4,15].

The computational domain used in the analysis is depicted in Fig. 2(c). A fixed heating power of 5 W is applied to the bottom surface of the heat sink base, corresponding to a heat flux of $12,500 \text{ W/m}^2$. Across all simulated conditions, the heat sink temperature varies between 300 K and 350 K. Within the considered temperature range, the thermophysical properties of the materials remain nearly constant, exerting minimal influence on the computational results. To simplify the numerical modeling, all material properties are assumed to be constant and are evaluated at 300 K. The heat sink is composed of Al6061, characterized by the following properties: density (ρ) = $2,719 \text{ kg/m}^3$, thermal conductivity (k) = 167 W/m-K , and specific heat capacity (C_p) = 896 J/kg-K . Although the thermal conductivity of Al6061 varies by approximately 5 % between 300 K and 350 K, this degree of variation is generally considered negligible in similar computational studies. Furthermore, because the primary objective of this study is to compare the relative performance of different configurations under consistent conditions, the assumption of constant properties does not compromise the validity of the results. The properties of air within the computational domain are as follows: $\rho = 1.177 \text{ kg/m}^3$, $k = 0.0264 \text{ W/m-K}$, $C_p = 1,006.43 \text{ J/kg-K}$, $\mu = 1.845 \times 10^{-5} \text{ Pa s}$, and $\text{Pr} = 0.707$.

To ensure full boundary layer development and achieve steady-state conditions, the inlet section is extended to ten times the hydraulic diameter ($10D_h$), preventing flow reversal and improving simulation accuracy [15,21]. Similarly, the outlet is positioned at $10D_h$ to minimize backflow effects, allowing the pressure drop to stabilize and ensuring that the flow is fully developed with no reverse flow at the exit [15,19,26]. The numerical simulations impose a velocity inlet boundary condition at 300 K and a pressure outlet condition with a gauge pressure of zero. The external walls were assigned symmetrical boundary conditions, establishing the periodic structure. All walls, except for the fluid-solid interface, were designated as no-slip and adiabatic surfaces to ensure proper coupling between the inner fin surfaces and the fluid flow.

2.4. Grid generation and independent test

The computational domain for the conjugate forced convective heat transfer simulation of the channel heat sink was developed using Ansys software (Ansys Inc., ANSYS 2022 R2), incorporating both solid and fluid regions. A combination of hexahedral and tetrahedral elements was employed for meshing, with particular attention given to optimizing mesh parameters to maintain uniformity while accommodating geometric curvature and complexity. To accurately capture boundary conditions and conductive heat transfer, highly refined unstructured grids were applied in the fluid domain near the fin surfaces and within the solid domain of the fins. This comprehensive grid analysis confirmed that the simulation results are dependent on the grid resolution and ensured high-accuracy outcomes.

Grid independence was assessed to mitigate errors associated with coarse meshing. This was achieved by evaluating at least four different mesh configurations, ranging from coarse to fine, with a growth rate maintained below 1.05. Details of the grid generation are provided in Table 2. The meshed interface between the solid and fluid domains is depicted in Fig. 3(a). By increasing the grid count from 161,785 to 4,653,091, multiple simulations were performed to evaluate the influence of node and element density on the surface temperature of the fins and the maximum air velocity, as shown in Fig. 3(b). The results indicated that for a grid size of 883,309 elements, the percentage error in surface temperature compared to 278,939 and 4,653,091 elements were 0.57 % and 1.07 %, respectively. Similarly, the percentage error in maximum air velocity for the same grid size was 3.91 % and 0.49 % when compared to 278,939 and 4,653,091 elements, respectively. These findings indicate that further refinement beyond 883,309 elements yields negligible improvements in accuracy, particularly for surface temperature predictions, where errors remain below 1 %. Although the velocity difference between the coarsest and selected meshes was 3.91 %, it decreased to only 0.49 % when compared to the finest mesh. This demonstrates that the selected mesh provides an optimal balance between computational efficiency and accuracy in both thermal and flow field simulations.

2.5. Validation and turbulence model

Numerous studies have employed laminar or k- ϵ turbulence models to characterize heat transfer in plate-fin heat sinks [3,11–13,15, 19–23,25,27,28,34,35]. The laminar, k- ϵ , k- ω , k-kl- ω , and SST turbulence models [1,2,17,24,30–32] were evaluated and compared against the experimental data reported by [24]. This study examines temperature variations under different air velocities. Temperature measurements were obtained using a FLIR E75 thermal imaging camera, which has a specified maximum measurement uncertainty of

Table 2
Grid generations.

Case	Grid spacing (m)		Growth rate	Total number of grids
	Coarse grids	Fine grids		
1	0.001	0.00100	1.05	161,785
2	0.001	0.00075	1.05	278,939
3	0.001	0.00050	1.05	883,309
4	0.001	0.00025	1.05	4,653,091

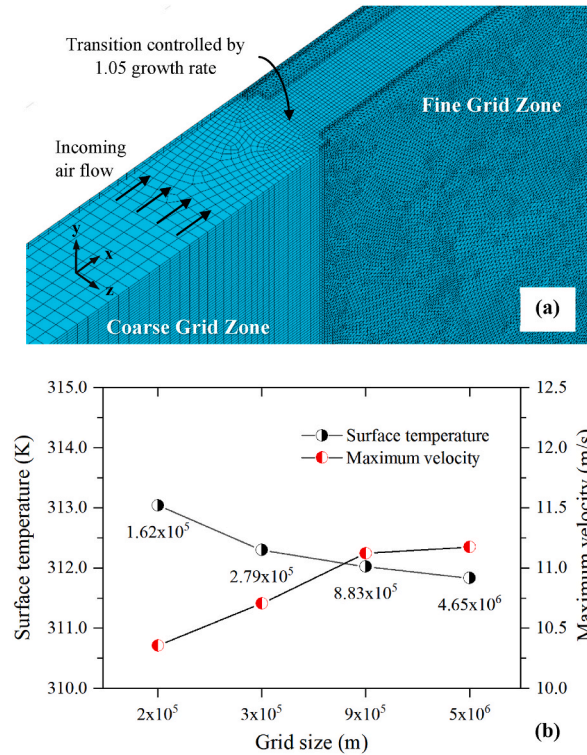


Fig. 3. Grid study results: (a) meshing configuration of the computational domain and (b) grid independence analysis.

± 2 %. For each experimental condition, temperatures were recorded at four equidistant locations along the fin surface at a consistent vertical height to assess measurement reproducibility. The variation among these points did not exceed 0.1°C , demonstrating high repeatability. The instrument's uncertainty was identified as the primary contributor to the overall measurement uncertainty. Therefore, the observed average deviation of 5.07 % between the simulation and experimental results is deemed acceptable and falls within the expected range of experimental error. The comparative results, presented in Fig. 4, indicate that the $k-\omega$ turbulence model exhibited the highest agreement with the experimental data, achieving an R^2 value of 0.998. This finding confirms the accuracy and reliability of the current conjugate forced convective heat transfer model, underscoring its suitability for future simulations of plate-fin heat sinks. Moreover, these validations confirm that the simulation results—including minor performance differences—are not artifacts of numerical errors or mesh sensitivity.

The $k-\omega$ turbulence model, particularly the Shear Stress Transport (SST) variant, is effective for simulating conjugate convective heat transfer in heat sinks operating within the laminar-to-turbulent transition regime ($Re \approx 5000$). Several studies have demonstrated the model's reliability in this transitional range. For instance, Rocha et al. [36] examined unsteady aerodynamics over a NACA0005 airfoil at Reynolds numbers of 1000, 2000, and 5000 using the $k-\omega$ SST model, successfully capturing key flow features such as vortex

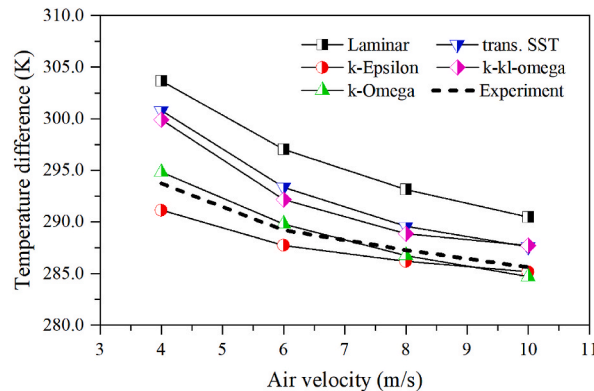


Fig. 4. Results of the conjugate forced convective heat transfer model analysis.

shedding and stall behavior. Similarly, Cruz et al. [37] performed both experimental and numerical analyses on single-phase pressure drop and heat transfer in helical corrugated tubes across Reynolds numbers ranging from approximately 300 to 5000. Their use of the $k-\omega$ SST model yielded results that aligned well with experimental data, confirming the model's capability to accurately simulate transitional flows in complex geometries. These findings collectively affirm the suitability of the $k-\omega$ SST turbulence model, particularly when enhanced with transition modeling techniques, for capturing the flow dynamics and heat transfer characteristics within the transitional Reynolds number range.

3. Data reduction

Data reduction methods were employed to evaluate the performance characteristics of heat sinks. The Reynolds number is used to define the hydraulic diameter in the heat sink models. To maintain consistency and simplify comparisons, both the hydraulic diameter and Reynolds number for all channels are calculated based on inlet conditions. The Reynolds number is expressed as follows [3]:

$$Re = \frac{\rho U_i D_h}{\mu} \quad (6)$$

where Re is Reynolds number, U_i is an inlet velocity, and D_h is the hydraulic diameter.

The Nusselt number is written as follows [3,4]:

$$Nu = \frac{h D_h}{k} \quad (7)$$

where Nu denotes the Nusselt number, h denotes the average heat transfer coefficient, and k denotes the thermal conductivity evaluated at the mean fluid temperature.

The average heat transfer coefficient is computed as follows:

$$h = \frac{Q}{A_c (T_w - T_f)} \quad (8)$$

where Q denotes the heat flux supplied to the heat sink base, A_c denotes the convection heat transfer area, T_w denotes the average temperature of the fluid-solid interface, and T_f denotes the mean fluid temperature.

The apparent friction factor can be written as follows [3]:

$$f = \frac{\Delta p D_h}{2 \rho L U_i^2} \quad (9)$$

$$\Delta p = p_i - p_o \quad (10)$$

where f denotes the friction factor, Δp denotes the pressure difference between the inlet and outlet of the heat sink channel, L denotes the length of the heat sink, p_i denotes the inlet pressure, and p_o denotes the outlet pressure.

The thermal enhancement factor also referred to as the heat transfer enhancement factor, is defined as the ratio of the overall heat transfer coefficient of a proposed channel to that of a conventional or original channel under constant input power. This relationship is expressed as follows [3,15]:

$$\eta = (\xi_{Nu}) (\xi_f)^{-\frac{1}{3}} \quad (11)$$

$$\xi_{Nu} = \frac{Nu}{Nu_o} \quad (12)$$

$$\xi_f = \frac{f}{f_o} \quad (13)$$

where η denotes the thermal enhancement factor, ξ_{Nu} denotes the relative Nusselt number, ξ_f denotes the relative friction factor, Nu_o is the Nusselt number of the original design, and f_o is the friction factor of the original design.

Equation (11) employs a performance evaluation criterion widely adopted in the literature, in which the exponent of $(-1/3)$ captures the cubic relationship between pressure drop and the required pumping power. This formulation has been validated by several studies as an effective means of balancing thermal enhancement and flow resistance in heat sink design. Although more rigorous methods, such as Entropy Generation Minimization (EGM), offer deeper thermodynamic insight, the present approach serves as a practical and widely accepted metric for comparative performance evaluation [38-40].

4. Results and discussion

4.1. Temperature distribution

The low-temperature airflow entered the heat sink channel at a Reynolds number of 500. Heat was transferred from the heated base of the heat sink to the airflow inside the channel and the plate fins through convection and conduction, respectively. This process represents conjugate forced convective heat transfer, which is visualized using a color contour in Fig. 5. In the contour representation, high temperatures are indicated by light yellow, while low temperatures are shown in blue. A significant heat transfer rate was observed within the channel airflow. As the airflow exited the channel, its temperature increased due to heat transfer, causing its color in the visualization to shift toward red.

The conjugate forced convective heat transfer was simulated for various channel heat sink designs at Reynolds numbers ranging from 500 to 5000. The mean fluid temperature, defined as the average air temperature over the normal plane located 0.01 m in front of and behind the heat sink, was analyzed and compared for each design at different Reynolds numbers. The results are illustrated in the graphs in Fig. 6(a). The results indicate that the average temperature within the proposed plate-fin designs was consistently lower than that of the conventional design across all Reynolds numbers. This finding demonstrates the effectiveness of the proposed designs in enhancing conjugate forced convective heat transfer. Additionally, these designs contribute to improving the thermal efficiency of the plate-fin heat sink.

The heat sink temperature (solid domain) simulated at a Reynolds number of 500 is illustrated in Fig. 6(b). It can be observed that the lower temperature contour is located at the front of the heat sink, while the higher temperature is found at the rear. The lower temperature is also observed in the upper part of the heat sink due to the heat flux applied to the heat sink base. All simulation configurations in this study exhibit this pattern.

4.2. Characteristics of fluid flow

Fig. 7 illustrates the airflow inside the heat sink channels at a Reynolds number of 3000 using color contours and streamlines. High velocity regions are represented in red, while low velocity areas are shown in blue. The airflow velocity contours exhibit a similar color gradient across all heat sink designs. However, the conventional channel heat sink displays a larger red region compared to the proposed designs, indicating higher velocity in certain areas. Regions with denser streamlines correspond to higher velocity, suggesting that the fluid moves faster in those areas. Notably, the conventional channel heat sink exhibits denser streamlines, indicating increased airflow velocity. The narrow-coupled channels in the heat sink function as a nozzle, further accelerating the airflow within the channel.

Fig. 8 presents the velocity distribution along the length of the heat sink at a height of 0.025 m from the heat sink base, allowing for a comparison among different heat sink designs. The results indicate that velocity is highest at the outlet ($x = 0.05$ m). However, in the case of the conventional plate-fin heat sink, velocity is also high at the inlet ($x = 0.00$ m). Additionally, the velocity along the length of the conventional plate-fin heat sink remains higher than that of the other heat sinks. In contrast, the fillet plate-fin heat sink exhibits the lowest velocity values throughout its length, with an outlet velocity comparable to that of the conventional heat sink. The lower

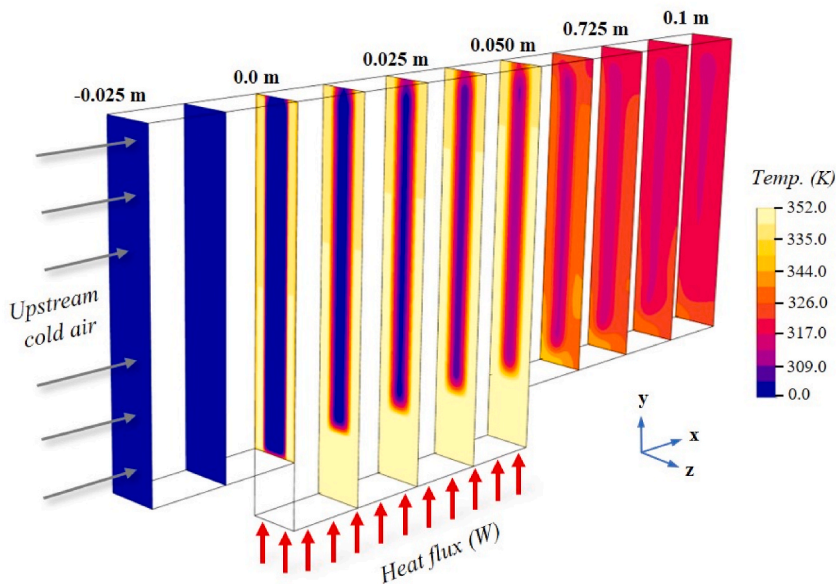


Fig. 5. Temperature distribution contours on slice planes within both the solid and fluid domains for a Reynolds number of 500.

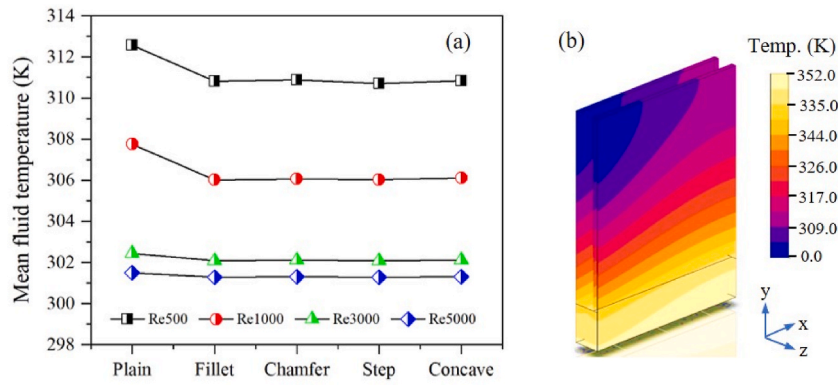


Fig. 6. Temperature distribution in the fins: (a) comparison of the average temperature for each plate-fin heat sink and (b) color contour of the conventional plate-fin heat sink.

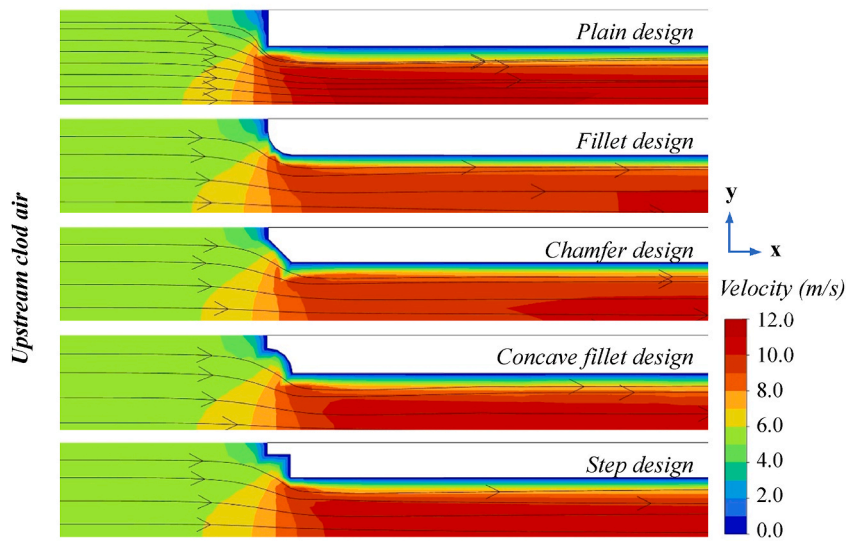


Fig. 7. Velocity contour with streamlines on half of the fluid domain at 0.025 m plane above the heat sink base for $Re = 3000$ under various inlet heat sink configurations.

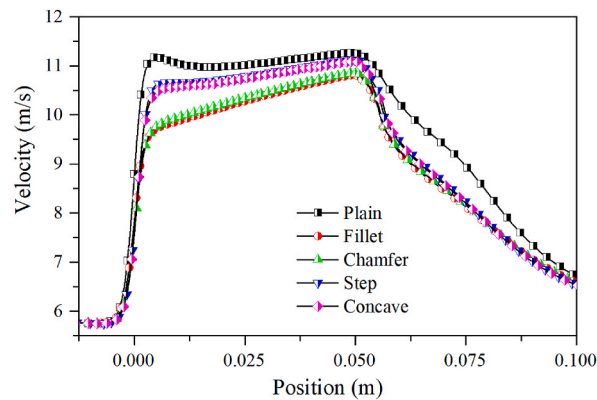


Fig. 8. Comparison of axial velocity profiles at a plane 0.025 m above the heat sink base for $Re = 3000$.

inlet velocity values in all heat sinks contribute to an increase in the different pressure values of the inlet and outlet.

4.3. Characteristics of pressure gradient

The pressure gradient in a plane located 0.025 m above the base of the heat sink is depicted using a blue contour, as shown in Fig. 9. The pressure gradient contours remained consistent across Reynolds numbers ranging from 500 to 5000. The dark blue region, representing the maximum pressure, was observed at the inlet of the heat sinks. Notably, the conventional plate-fin heat sink exhibited a larger dark blue area at the inlet compared to the other plate-fin heat sinks, while the smallest dark blue region was observed at the inlet of the fillet and chamfer plate-fin heat sinks.

The pressure values at the midpoint of several fins, measured 0.025 m above the base along the length of the heat sink, are presented graphically for comparison in Fig. 10. The results indicate that the conventional plate-fin heat sink exhibited the lowest pressure values along its length, whereas the fillet and chamfer plate-fin heat sinks displayed the highest-pressure values. The pressure values at the outlet of each heat sink were comparable. Moreover, the pressure at the inlet of the heat sinks was generally higher than at the outlet, except for the conventional plate-fin heat sink, where the inlet pressure was nearly equal to the outlet pressure.

4.4. Characteristics of heat transfer

The Nusselt number and friction factor for each heat sink at a Reynolds number of 3000 were calculated and are presented graphically for comparison in Fig. 11. The conventional plate-fin heat sink serves as the benchmark, represented by a dashed line in the graphs. The results indicate that the Nusselt number for the fillet and chamfer plate-fin heat sinks exceeded the benchmark, whereas the step plate-fin heat sink exhibited a lower Nusselt number, and the concave fillet plate-fin heat sink closely approximated the benchmark. In contrast, the friction factor for all proposed heat sinks was lower than that of the benchmark. The fillet and chamfer plate-fin heat sinks demonstrated a relative Nusselt number greater than 1.00, while the step and concave fillet plate-fin heat sinks had a relative Nusselt number below 1.00. Nonetheless, the relative friction factor for all proposed plate-fin heat sinks remained below 1.00. This outcome underscores that the friction factor effectively captures relative changes in flow resistance, even when the absolute pressure drop is minimal. The observed reduction in the friction factor, despite the low-pressure differential, indicates significant improvements in flow efficiency resulting from the optimized channel geometries.

4.5. Comprehensive performance

The thermal enhancement factor of each proposed plate-fin heat sink was calculated for Reynolds numbers ranging from 500 to 5000 and is presented graphically in Fig. 12. The results indicate that the thermal enhancement factor increased with increasing Reynolds number, demonstrating higher values at elevated Reynolds numbers. All proposed plate-fin heat sinks exhibited a thermal enhancement factor greater than 1.00, except for the step plate-fin heat sink. These findings indicate that incorporating fillet and chamfer features alters the flow structure by mitigating separation caused by sharp edges. This modification leads to an increase in the Nusselt number and a reduction in the friction factor, thereby improving the overall thermal enhancement efficiency of the modified heat sink.

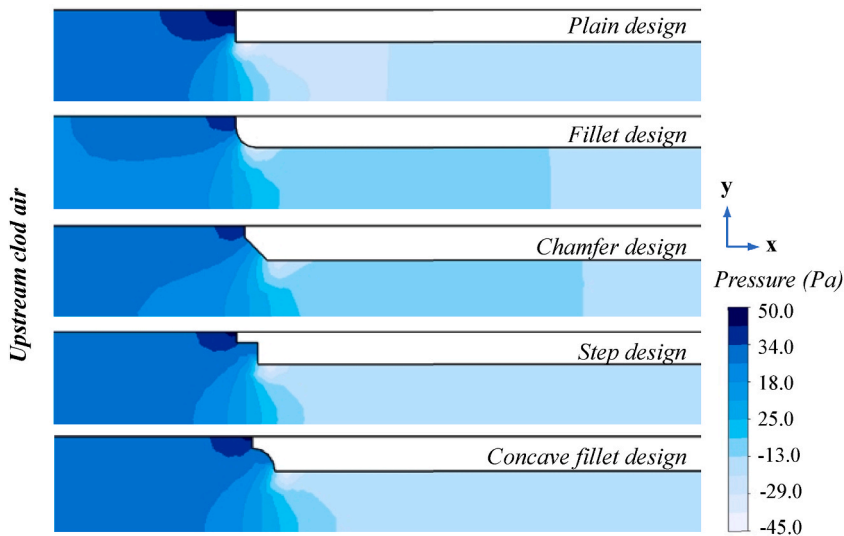


Fig. 9. Pressure contour with streamlines for half of the fluid domain at a plane 0.025 m above the heat sink base for $Re = 3000$ under various inlet heat sink configurations.

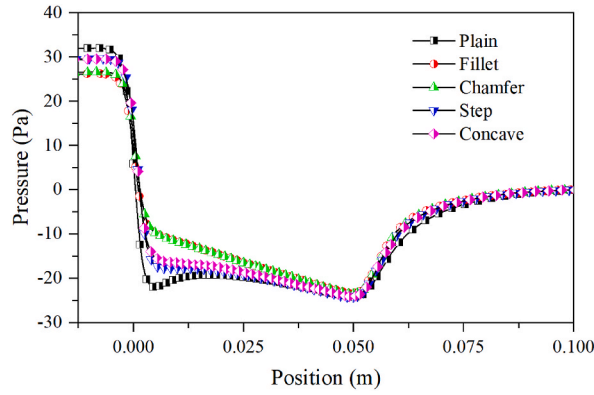


Fig. 10. Comparison of axial pressure profiles at a plane 0.025 m above the heat sink base for $Re = 3000$.

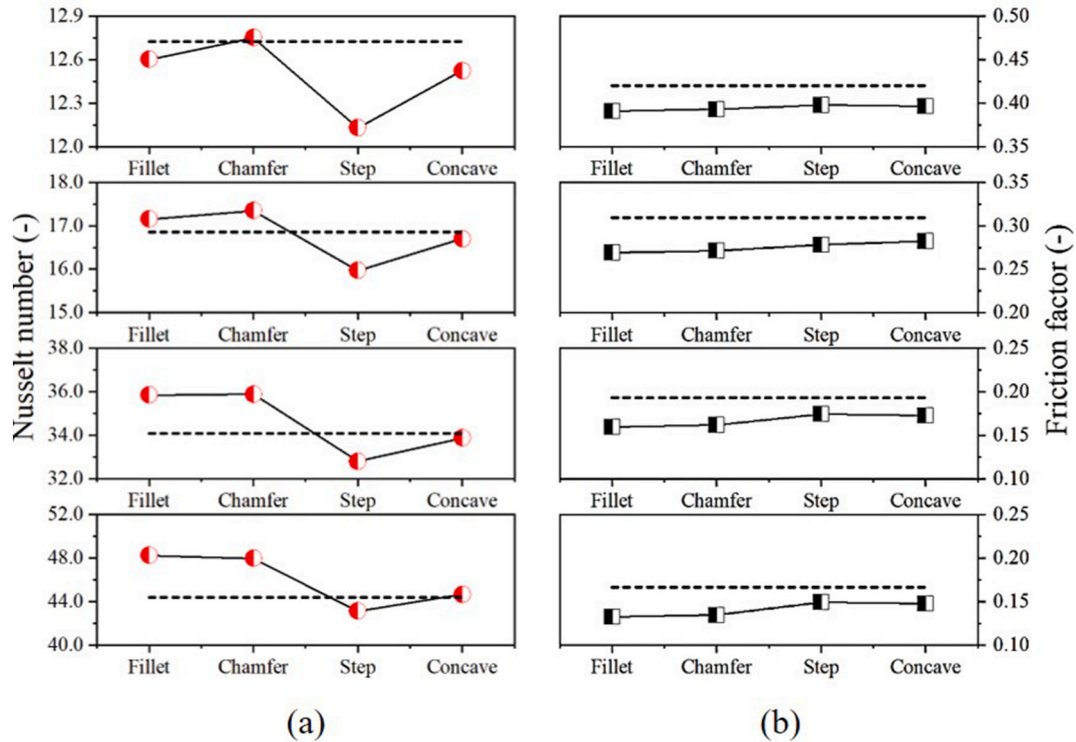


Fig. 11. Heat transfer performance across different heat sink configurations: (a) Nusselt number (Nu) comparison and (b) apparent friction factor (f). The plain heat sink is used as a reference and is indicated by the red dashed line. Results are presented for Re of 500, 1000, 3000, and 5000, arranged from top to bottom, respectively.

5. Optimization of the plate-fin configurations

The fillet and chamfer plate-fin heat sinks played a significant role in enhancing the thermal enhancement factor, increasing it from 1.01 to 1.17 and from 1.02 to 1.16, respectively, over a Reynolds number range of 500–5000 at the proposed fillet radius ($R1$) and chamfer length ($C1$). The conjugate forced convective heat transfer within the channel heat sink is primarily influenced by the thickness of the plate fins when calculating the thermal enhancement factor [41,42]. To determine the maximum thermal enhancement factor for each plate-fin configuration, the fillet radius and chamfer length were dimensionless to the fin thickness and varied. Their dimensions were expressed as ratios relative to the plate-fin thickness, denoted as t/R and t/C for the fillet and chamfer plate-fin heat sinks, respectively, with values ranging from 1.0 to 3.0. Simulations were conducted across a Reynolds number range of 500–5000, with the results presented in Fig. 13. The findings indicate that the maximum thermal enhancement factor for the fillet plate-fin heat sink was achieved at an t/R ratio of 2.0, while the chamfer plate-fin heat sink reached its peak enhancement at a t/C ratio

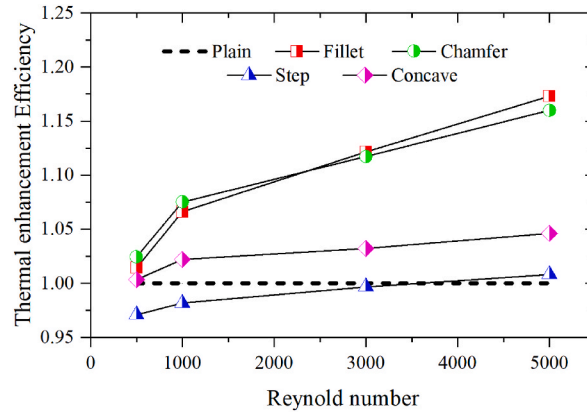
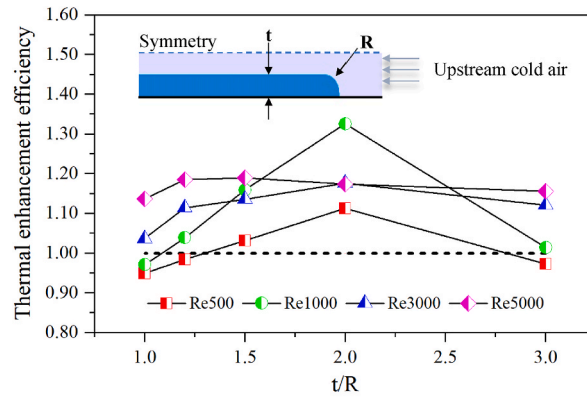
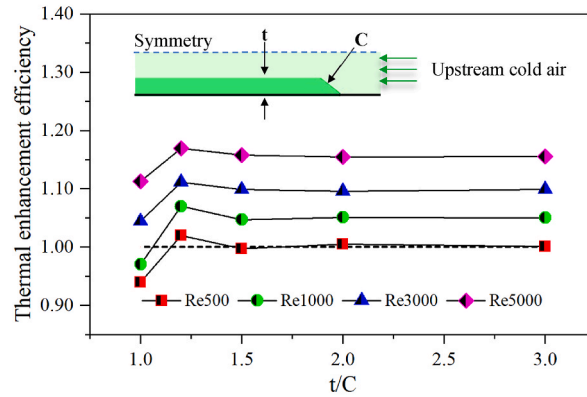


Fig. 12. Comparison of thermal enhancement efficiency across various heat sink configurations. The plain heat sink is indicated by the red dashed line for reference.



(a)



(b)

Fig. 13. the enhancement factors of the (a) fillet and (b) chamfer plate-fin heat sinks.

of 1.2. Moreover, the sensitivity analysis indicates that the optimized design is robust and not overly sensitive to parameter variation, as small deviations ($\pm 10\%$) in t/R and t/C only slightly alter thermal enhancement efficiency. This demonstrates the suggested configuration's dependability in practical situations. These optimized ratios provide essential guidelines for the future manufacturing of the proposed plate-fin heat sink configurations.

6. Conclusion

This study examined the thermal performance enhancement of plate-fin heat sinks, with a particular focus on maximizing the Nusselt number and minimizing the friction factor to improve the overall thermal enhancement factor. The results underscore the importance of enhancing convective heat transfer while reducing the temperature differential between the cooling airflow and the plate fins. Moreover, the observed low pressure drop across the channel contributed to a reduced friction factor, further supporting efficient thermal performance.

Among the various computational fluid dynamics (CFD) approaches evaluated, the conjugate forced convective heat transfer model emerged as the most effective for accurately analyzing and optimizing heat sink configurations. In parallel, the SST $k-\omega$ turbulence model demonstrated superior predictive capability for capturing flow behavior and heat transfer characteristics within the domain.

While contemporary trends in heat sink design often emphasize intricate fin geometries that can enhance performance, such designs typically entail higher manufacturing complexity and cost. In contrast, the present study introduces novel plate-fin configurations that balance thermal efficiency with manufacturability. Through the strategic application of simple geometric modifications—namely, fillets and chamfers—and the optimization of the t/R and t/C ratios to 2.0 and 1.2, respectively, the proposed designs achieve significant improvements in heat transfer without compromising ease of fabrication using conventional machining techniques.

Although a full-scale parametric optimization was beyond the scope of this work, the findings clearly demonstrate that practical geometric refinements can yield substantial gains in thermal performance. These insights provide a valuable foundation for future investigations aimed at further optimizing heat sink designs in both academic and industrial applications.

CRediT authorship contribution statement

Panit Kamma: Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Data curation, Conceptualization, Formal analysis, Software. **Kittipos Loksupapaiboon:** Writing – review & editing, Writing – original draft, Investigation, Formal analysis, Data curation, Conceptualization, Methodology, Validation. **Juthanee Phromjan:** Writing – review & editing, Writing – original draft, Investigation, Conceptualization, Formal analysis, Visualization. **Machimontorn Promtong:** Writing – review & editing, Writing – original draft, Software, Investigation, Formal analysis. **Chakrit Suvanjumrat:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Project administration, Formal analysis, Conceptualization, Data curation, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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